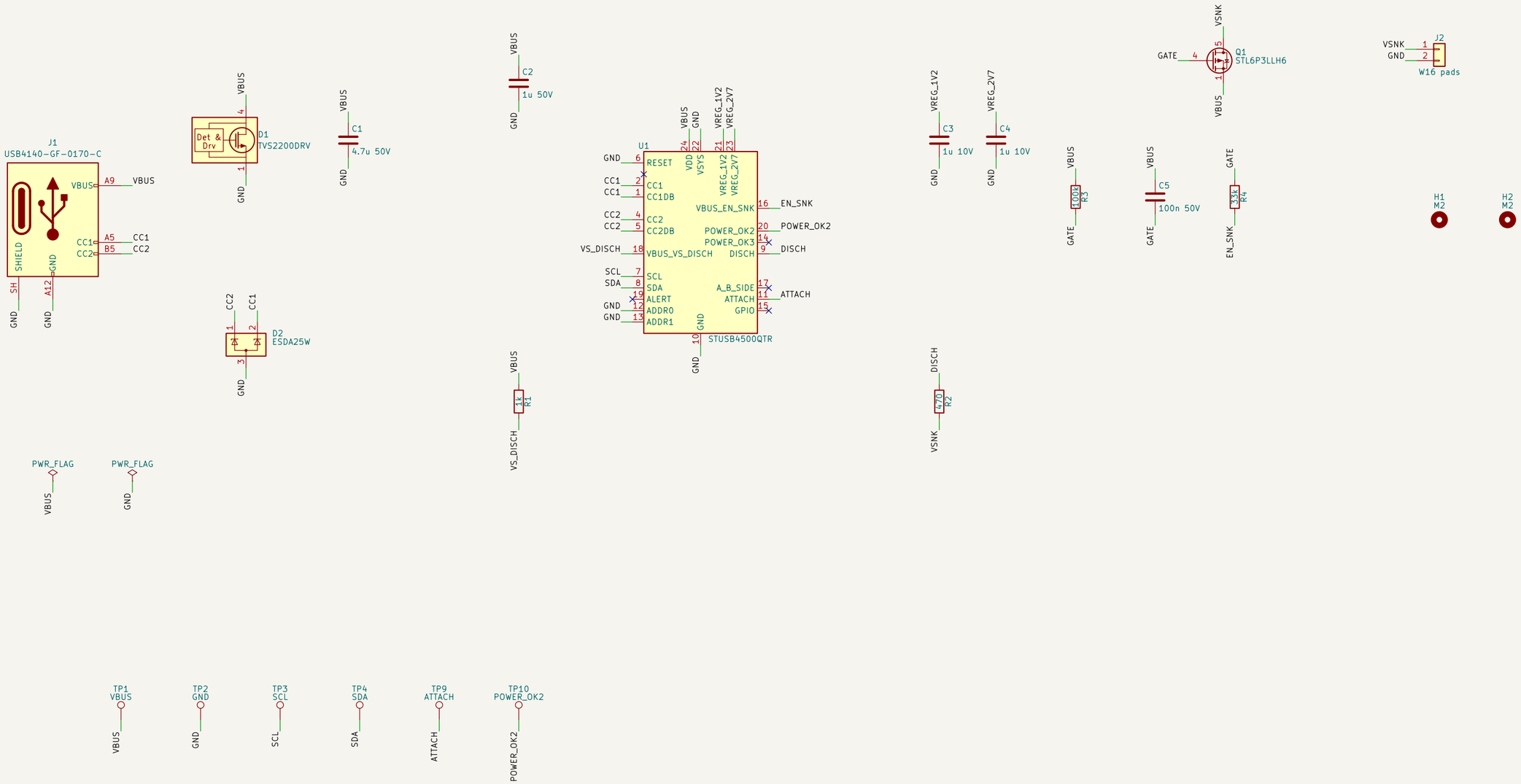


PCB-13 charge inlet, rev A. Spec: charge-path.md sec 3 (D-040), inputs frozen in D-069 (04-pcbs/pcb-13-charge-inlet/README.md).
J1 GCT USB4140-GF-0170-C on the -X (panel) face, hand-soldered by the builder (D-069): VBUS A9/B9, GND A12/B12 and the four shell stakes (SH), CC1 A5, CC2 B5.
D1 TVS2200 (22 V stand-off, 28.35 V max clamp) and C1 at J1; D2 ESDA25W on CC1 (pin 2) and CC2 (pin 1): identical common-anode diodes, swapped for routing. Everything on VBUS is rated >= 28 V.
U1 STUSB4500 auto-run sink, NVM 5/9/15 V at 1.5 A with POWER_ONLY_ABOVE_5V (sec 8). Dead battery: CC1DB-CC1, CC2DB-CC2. VSYS to GND (unused).
Q1 STL6P3LLH6: S on VBUS, D on VSNK; body diode blocks VSNK -> VBUS before a contract. R3 100k / R4 33k: VGS = -0.75 x VBUS. C5: tau 2.5 ms soft start.
R1 1k 0805 anti-surge on VBUS_VS_DISCH (20 mA, 0.4 W pulses at 20 V); R2 470 R 1206 anti-surge DISCH -> VSNK (32 mA at 15 V).
ADDR0, ADDR1, RESET tied to GND at the QFN (address 0x28; ST: to ground with no MCU; placement, D-069 amended). ALERT, GPIO, POWER_OK3, A_B_SIDE, NC: no connection.
TP1-TP4 programming (VBUS, GND, SCL, SDA) on a 2.54 mm grid for P75 pogos; TP9, TP10 probing (ATTACH, POWER_OK2); VSNK on J2 pad 1, CC on D2's leads. SCL/SDA pull-ups (3.3 V) are on the jig.
J2: W16 solder pads, pin 1 VSNK, pin 2 GND, to PCB-02 J2-8 (pin 1 VSNK, pin 2 GND). H1, H2: 2.2 mm holes at +-11 mm for the M2 x 5 screws (BO-017).

Spec: project/03-build/04-pcbs/charge-path.md sec 3; inputs: pcb-13-charge-inlet/README.md
USB-C PD sink: GCT USB4140, STUSB4500, TVS2200, ESDA25W, STL6P3LLH6 (D-040, D-069)

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Title: PCB-13 charge inlet

Size: A3	Date: 2026-10-29	**Rev: A**
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